

Visualization Tool for Electric Vehicle Charge and Range Analysis
using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	3 MARKS

6.3 – Data Preparation

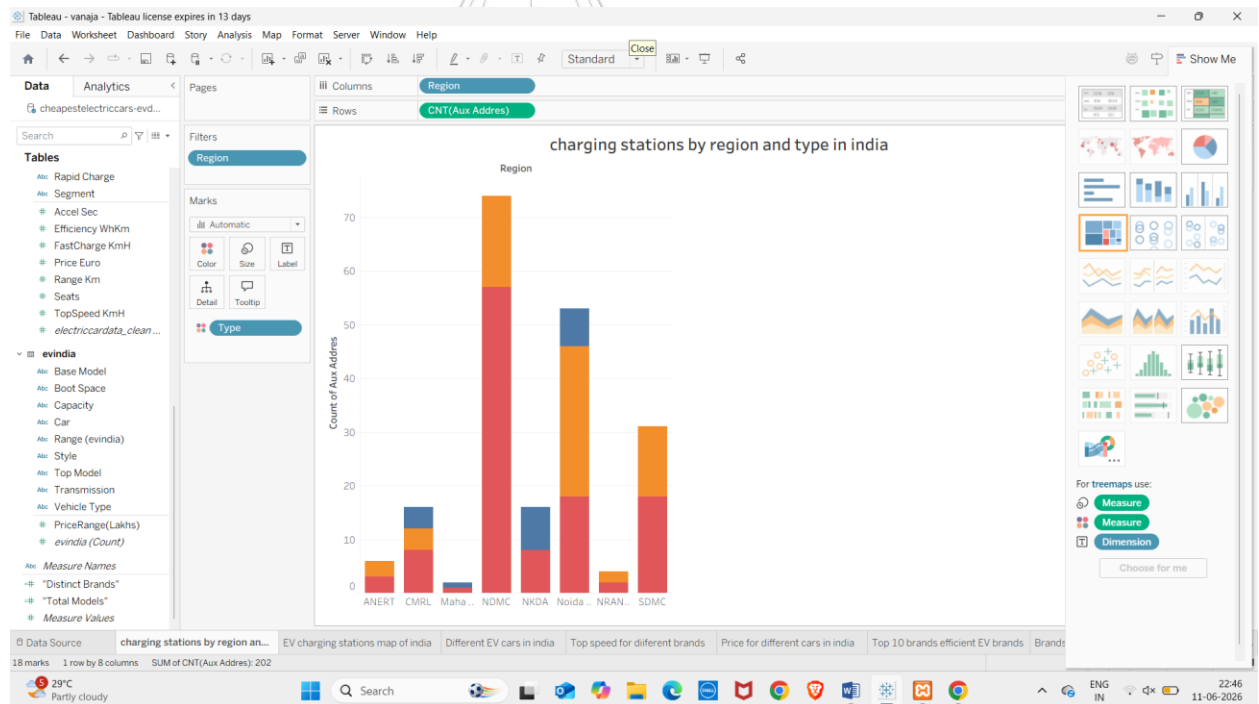
SHEETS	TYPE
Charging stations by region and type in India	Bar chart
EV charging stations map of India	Geographical map
Different EV cars in India	Text table
Top speed for different brands	Statistical box
Price for different cars in India	Scatter plot
Top 10 most efficient EV brands	Horizontal bar
Brands according to body style	Heat map
Brands filtered by powertrain type	Packed bubble
Number of models by each brand	Tree map
Summary card for different brands of EV cars globally	KPI summary cards
Summary card for different brands of EV cars in India	KPI summary cards

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Sheet-01: Charging Stations by region and type in India

- Drag Region to the Columns Shelf.
- Drag Aux Address to the Rows Shelf.
- Now, change Aux Address into count (Aux Address)
- Drag Type to Colour on the Marks card.
- Now, Drag region on Filters
- On the Marks card, select Bar from the drop-down menu.

Output:



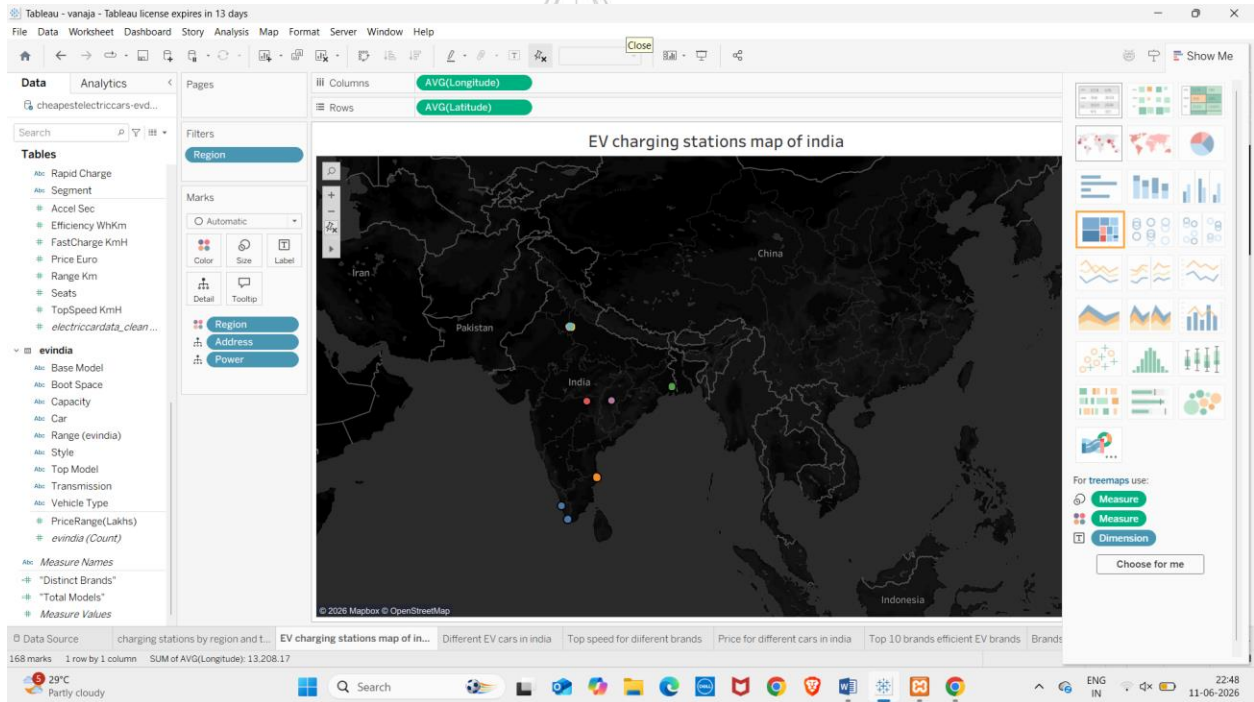
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Sheet-02: EV Charging stations map of India

- Drag Longitude to the column Shelf.
- Drag Latitude to the row shelf.
- Drag Region to Colour on the Marks card.
- Drag Address and Power to Details on the Marks card.

Output:



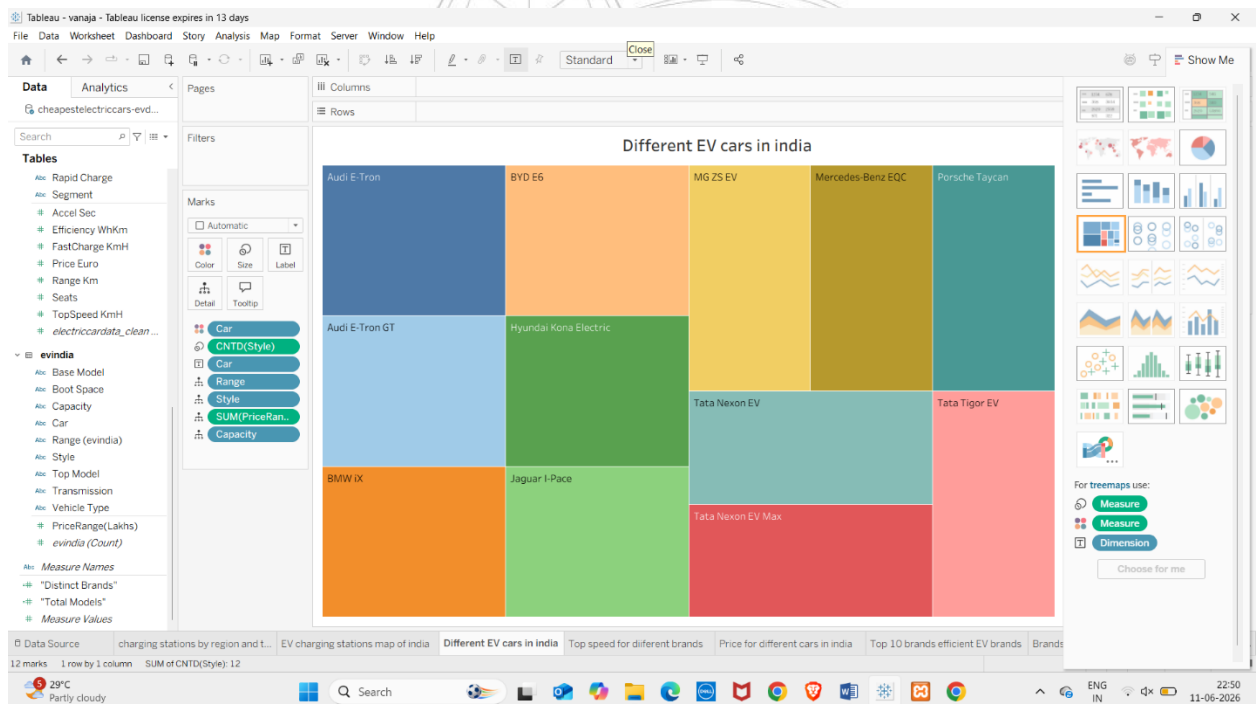
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Sheet-03: Different EV cars in India

- Drag Brand to the Rows Shelf.
- Drag Model to the Rows Shelf (to the right of Brand).
- Drag car to Text on the Marks card.
- Drag Style to size on the Marks card.
- Drag Range and Style to Details on the Marks card
- Then, Drag price Range(Lakhs) and capacity to Details on the Marks card
- Now, change price Range(Lakhs) into sum (price Range)

Output:



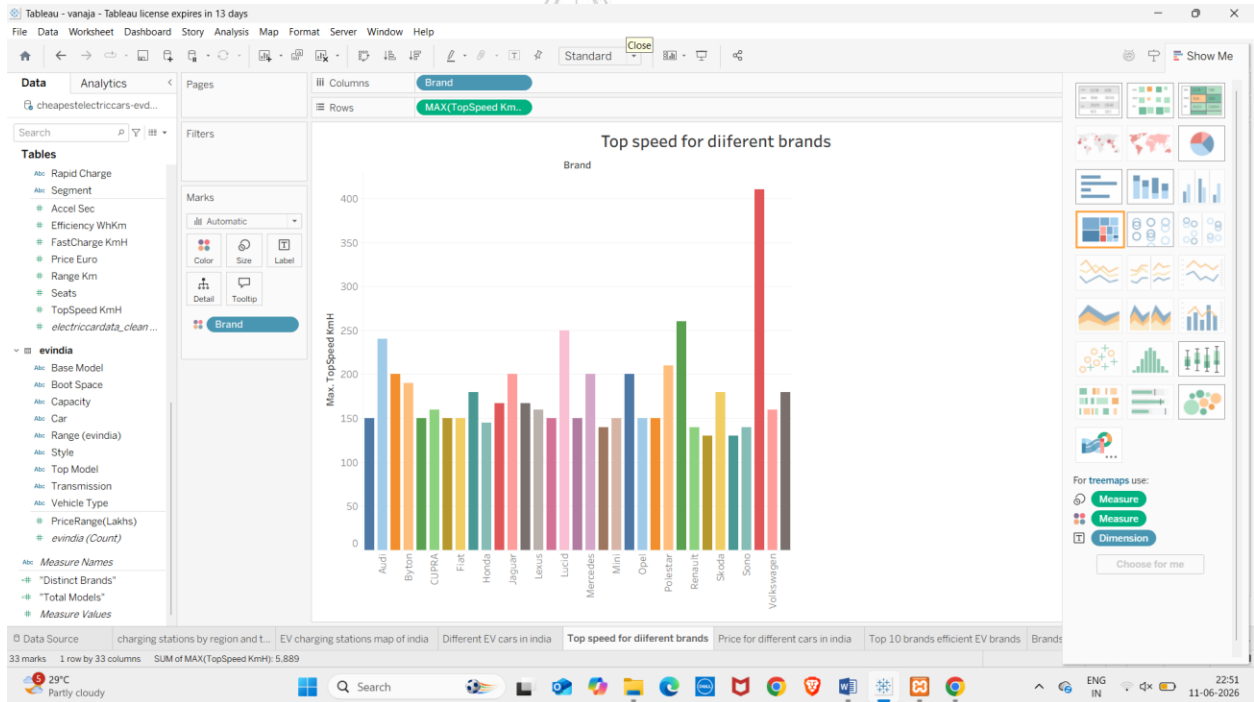
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Sheet-04: Top speed for different Brands

- Drag Brand to the Columns Shelf.
- Drag Top speed to the Rows Shelf.
- Right-click MAX (Top speed) on the Rows shelf, select Measure, and click Average.
- Drag Brand to the colour on the Marks card Top speed.

Output:



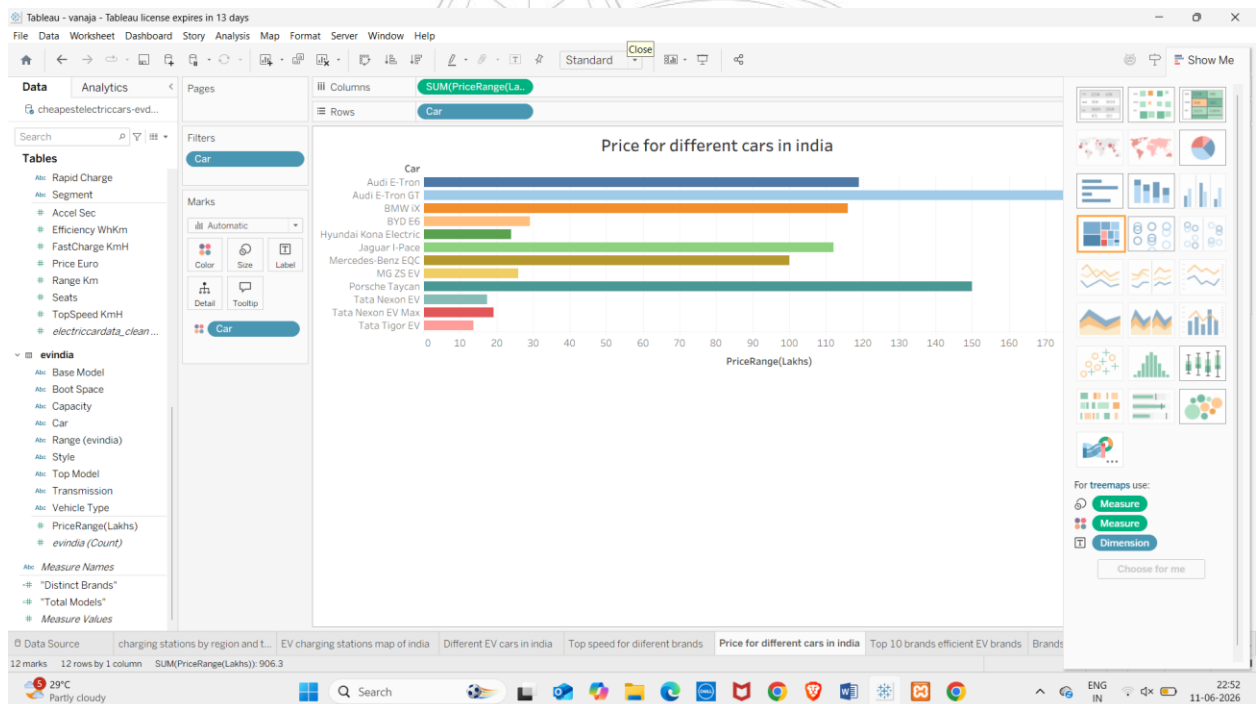
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Sheet-05: Price for different cars in India

- Drag price Range(Lakhs)to the Columns Shelf.
- Right-click SUM (price Range(Lakhs)) on the Rows shelf, select Measure, and click Average.
- Drag car to the Rows Shelf.
- Drag car to Colour on the Marks card.
- Drag car to Filters and select all.
- Then, click ok.

Output:



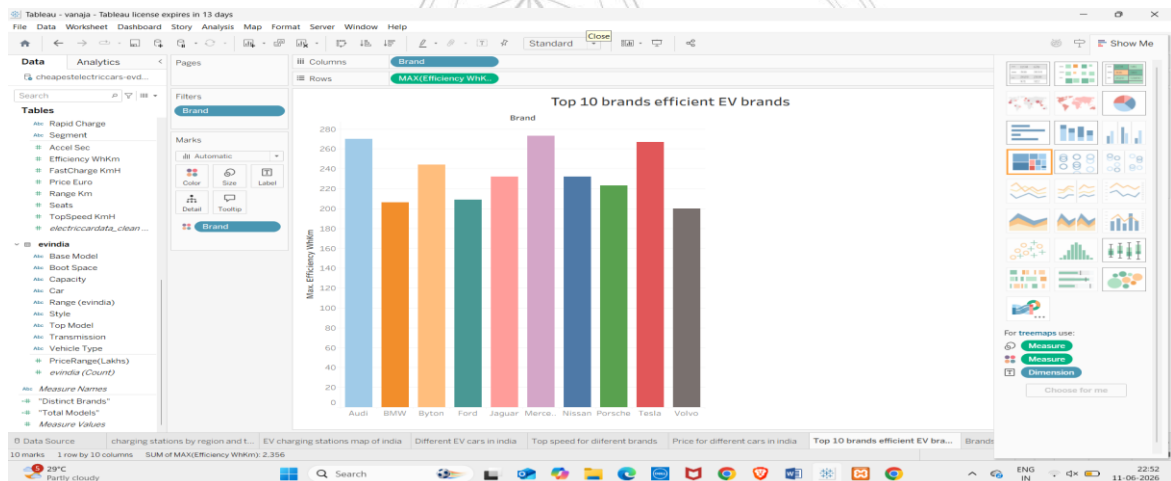
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Sheet-06: Top 10 most efficient EV Brands

- Drag Brand to the Rows Shelf.
- Drag EfficiencyWhkm to the Columns Shelf.
- Right-click SUM (EfficiencyWhkm) on the Columns shelf, select Measure, and click Sum.
- Drag Brand to the Filters Shelf, click the Top tab, select by Field, and set it to Top 10 by Efficiency (Average).
- Click the Sort Descending icon on the top toolbar.

Output:



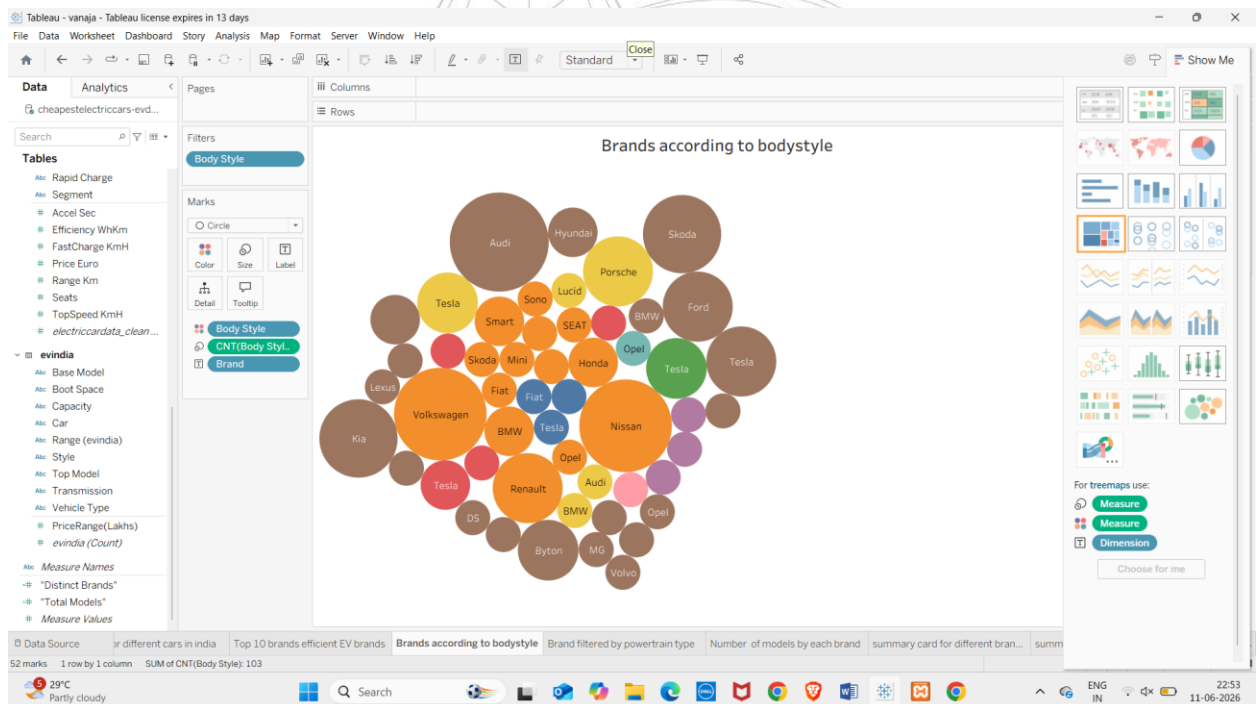
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Sheet-07: Brands according to Body style

- Drag Body Style to the Columns Shelf.
- Drag Brand to the Rows Shelf.
- On the Marks card, select circle from the drop-down menu.
- Drag Body Style to Colour on the Marks card,
- Drag Body Style to Size on the Marks card, right-click it, select Measure, and click Count.
- Drag Brand to the Label.

OUTPUT:



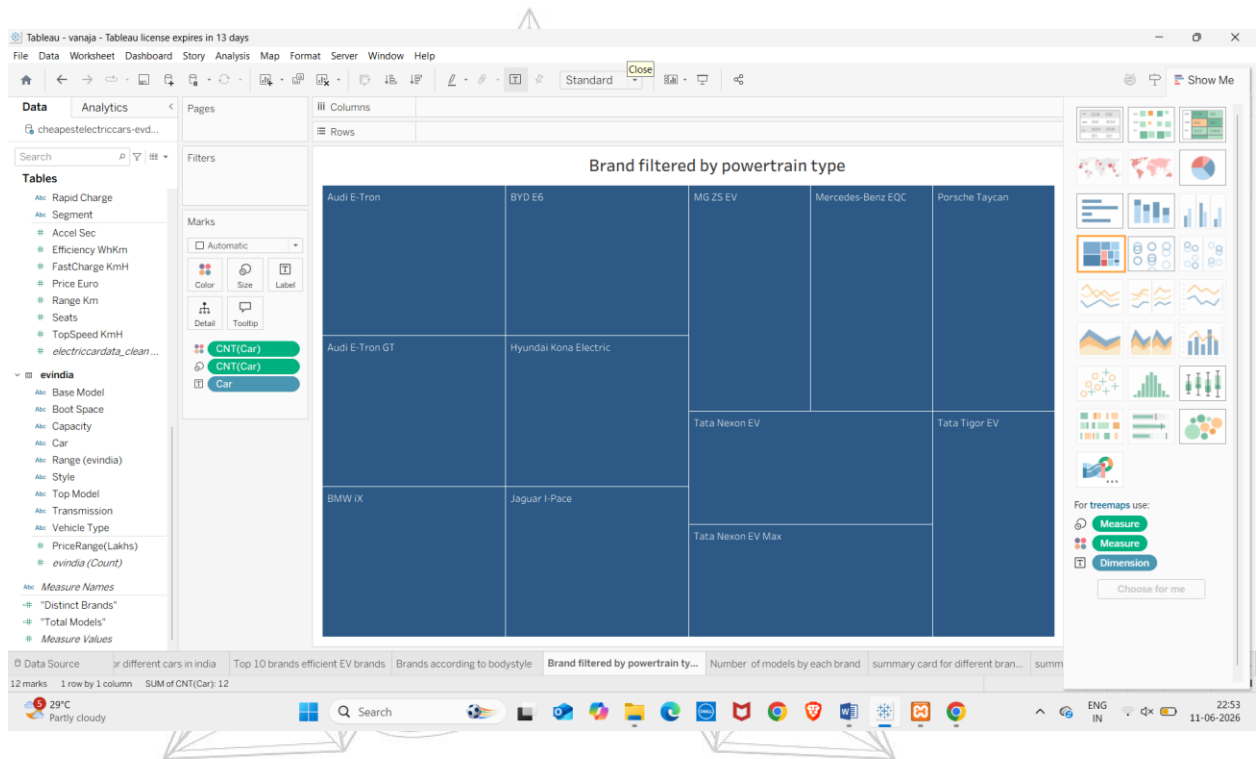
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Sheet-o8: Brand filtered by Powertrain type

- Drag car to Label on the Marks card.
- Drag Car to Size on the Marks card, right-click it, select Measure, and click Count.
- Drag Car to the colour, Right-click it, select Measure, and click Count.

OUTPUT:



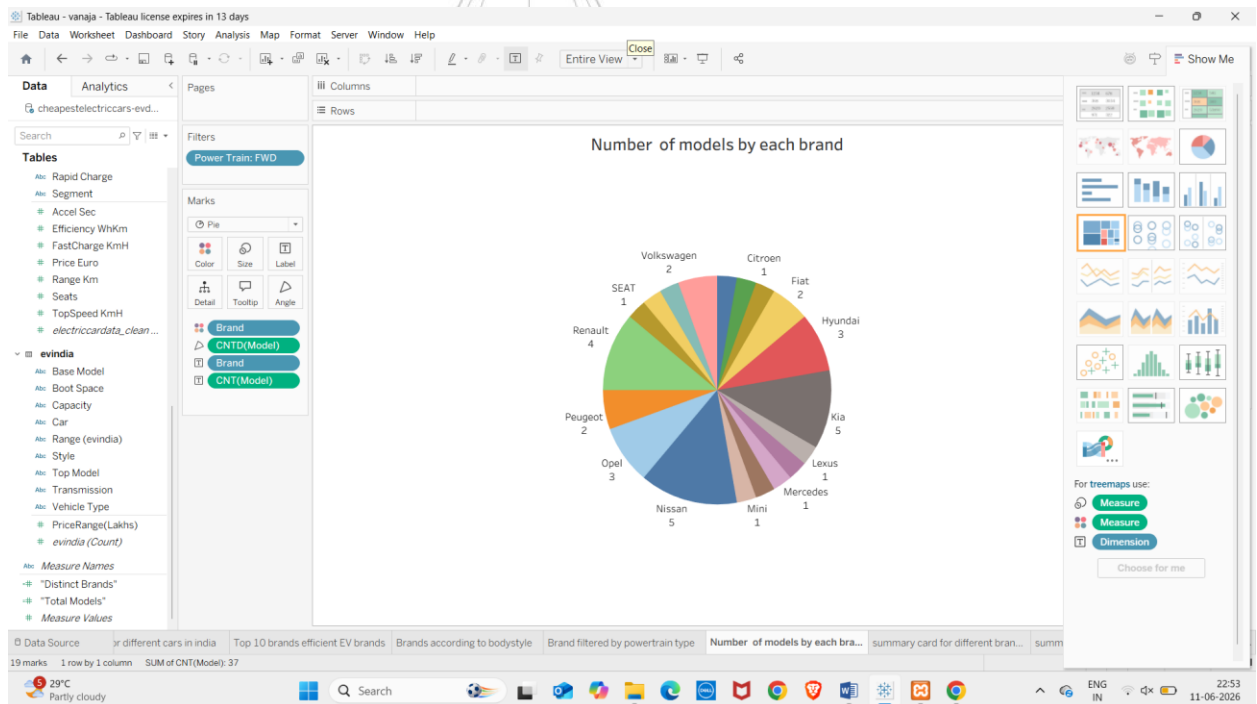
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Sheet-09: No of models by each brand

- Drag Brand to colour on the Marks card
- Drag Brand to Label on the Marks card.
- Drag Model to the Angle. Right-click Model on the Marks card, select Measure, and click Count.
- Drag Model to the Label. Right-click Model on the Marks card, select Measure, and click Count

OUTPUT:



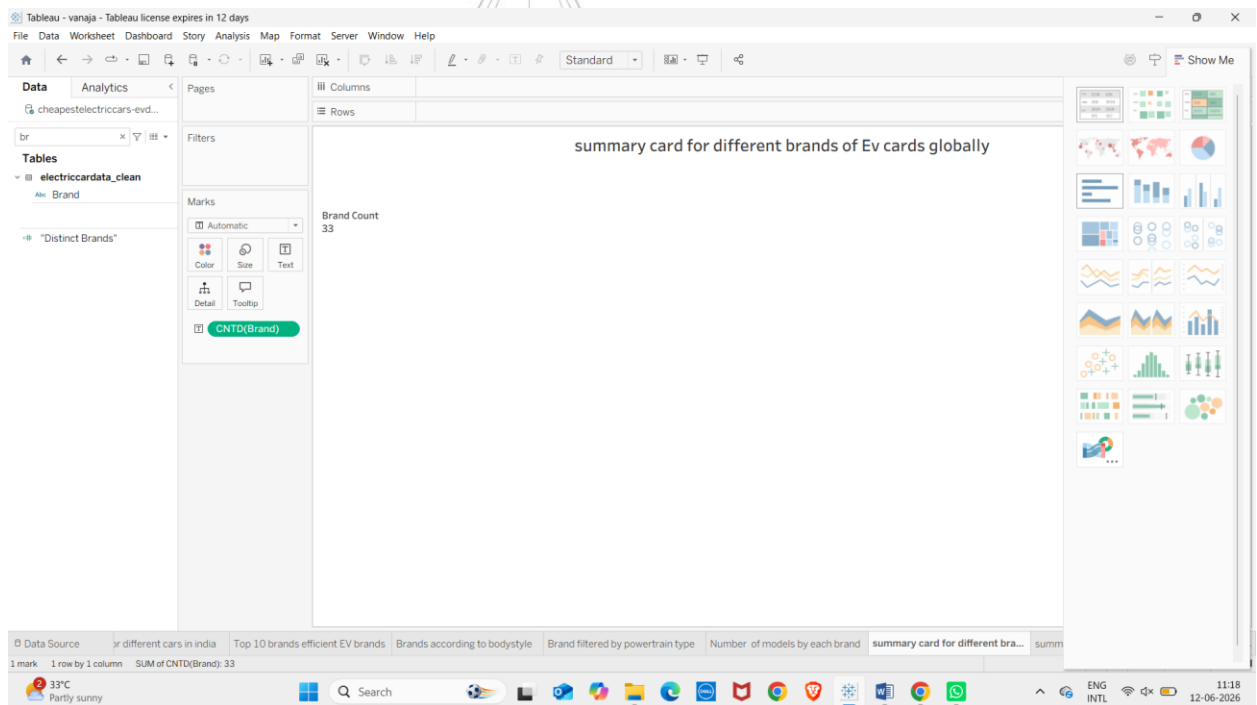
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Sheet-10: Summary card for Different brands of EV Cars globally

- Go to the top Analysis menu and click Create Calculated Field.
- Name it "Distinct Brands" and write the formula: COUNTD([Brand]).
- Create another Calculated Field, name it "Total Models", and write the formula: COUNT([Model Name]).
- Drag both new calculated fields to Text on the Marks card.

OUTPUT:



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